

Proposed Solutions-Template

Date	01 July 2026
Team ID	SWTID-2026-2152
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

1. Architectural Blueprint (The Solution Framework)

To transition from standard, static learning management systems to a responsive, human-centered ecosystem, the proposed solution implements a modular, decoupled architecture built on Streamlit and Python-driven AI microservices. By combining structural session handling with persistent model memory caching, the system evaluates student inputs and delivers contextual guidance seamlessly, maintaining a target end-to-end processing threshold of $<500\text{ms}$.

2. Core Functional Pillars of the Proposed Solution

Pillar 1: Hybrid Text-Based Emotion Analytics Pipeline

Instead of relying on single-metric classification, the solution deploys a dual-engine architecture to evaluate text inputs (such as question responses, chat logs, and reflections):

- **Sequential Baseline Tracking:** A Bidirectional Long Short-Term Memory (BiLSTM) classifier parses text strings into a 5-Class Softmax distribution to track immediate emotional states.
- **Contextual Nuance Extraction:** A parallel, fine-tuned BERT Classifier with custom class-weight adjustments extracts deep emotional intent, eliminating semantic blind spots like sarcasm or mixed sentiments.
- **Unified Prediction Schema:** A custom verification threshold detects secondary sentiments scaling $\geq 15\%$, allowing the system to identify complex, multi-layered student states (e.g., highly frustrated yet actively engaged).

Pillar 2: Field-Aware Generative AI Mentoring Engine

To ensure that feedback is both empathetic and accurate, the platform integrates a dynamic prompt generation layer wrapped around the Google Gemini API:

- **Contextual Inputs:** The engine programmatically binds the *Academic Field*, *Specific Problem Input*, *Detected Emotion*, and *Model Confidence Score* into a structured prompt.
- **Empathetic Interventions:** The system interprets this data to generate targeted hints, pacing adjustments, or motivational encouragement

tailored to the student's exact learning situation.

- **Fail-Safe Redundancy:** To prevent network or API rate-limiting issues from disrupting the learning flow, the architecture includes automated, programmatic fallback response templates.

Pillar 3: Stateful User Experience & Historical Analytics

Data transparency is achieved through a centralized frontend dashboard that bridges the gap between learner struggles and instructor tracking:

- **Session State Optimization:** Streamlit's native `st.session_state` parameters manage variables dynamically across form controls, text fields, and clear triggers without losing the active context.
- **Visual Confidence Metrics:** Interactive Plotly Charts and analytics tabs show model probability balances, helping users visualize their performance trends clearly.
- **Continuous Learning Logs:** All interaction histories, texts, and model outputs are written to a localized CSV Persistence Engine to support long-term tracking and offline trend analysis.

3. Solution Feasibility & Impact Mapping

Proposed Solution Module	Core Technical Component	Institutional & Pedagogical Value
Real-time Emotion Parsing Engine	Pre-loaded BiLSTM (5-Class) + Fine-Tuned BERT Pipeline.	Uncovers hidden learning bottlenecks by converting qualitative text struggles into quantifiable data points.
Context-Aware Adaptation Engine	Google Gemini API + Automated Template Fallbacks.	Provides a human-like, adaptive virtual mentoring experience that reduces drop-out rates and keeps students engaged.
Interactive Analytics Platform	Streamlit UI + Plotly Caching Engine + CSV Logging.	Equips students with personalized progress tracking while giving instructors real-time visibility to support

Proposed Solution Module	Core Technical Component	Institutional & Pedagogical Value
		struggling learners.